Connectivness of Graphs

Def. 1. *Vertices $u, v \in V, u \neq v$ are called* ***connected*** *if there is a $u - v$-path in G.*

Def. 2. *A graph $G = (V, E)$ is* ***connected*** *if $\forall u, v \in V, u \neq v$ a $u - v$-path in G exists.*

Def. 3. *Vertices $u, v \in V, u \neq v$ are called* ***k-connected*** *if $|V| \geq k + 1$ and*

$$\forall X \subseteq V \setminus \{u, v\}$$

$$\text{with } |X| < k,$$

$$u \text{ and } v \text{ are connected in } G[V \setminus X]$$

Def. 4. *A graph $G = (V, E)$ is called* ***k-connected*** *if $|V| > k$ and*

$$\forall X \subseteq V \text{ with } |X| < k$$

$$G[V \setminus X] \text{ is connected}$$

Def. 5. *$X \subseteq V \setminus \{u, v\}$ is called a $u - v$-* ***separator*** *if u and v are in different connected components of $G[V \setminus X]$.*

Rem. 6. *If G contains a vertex v with $deg(v) < k$, then G cannot be k-connected.*

Thm. 7. ***Menger's Theorem (Vertex Version)****: The minimum size of a $u - v$-separator is equal to the maximum number of internally vertex-disjoint $u - v$-paths. A graph is k-connected iff $\forall u \neq v$ there are k internally vertex-disjoint $u - v$-paths.*

Def. 8. *A graph $G = (V, E)$ is called* ***k-edge-connected*** *if*

$$\forall X \subseteq E \text{ with } |X| < k$$

$$(V, E \setminus X) \text{ is connected}$$

Thm. 9. ***Menger's Theorem (Edge Version)****: A graph G is k-edge-connected iff. $\forall u, v \in V, u \neq v$ there are k edge-disjoint $u - v$-paths.*

Thm. 10. *It always holds that:*

$$\text{Vertex Connectivity} \leq \text{Edge Connectivity}$$

$$\leq \text{Minimum Degree } \delta(G)$$

Lets look at the special case where $k = 1$:

Def. 11. *For a connected graph $G = (V, E)$:*

- *Vertex $v \in V$ is a* ***cut vertex*** *iff. $G[V \setminus \{v\}]$ is disconnected.*
- *Edge $e \in E$ is a* ***bridge*** *iff. $(V, E \setminus \{e\})$ is disconnected.*

Lem. 12. *If $e = \{x, y\} \in E$ is a bridge, then $deg(x) = 1$ or x is a cut vertex.*

Def. 13. ***Blocks****: Define an equivalence relation $\sim$ on E:*

$$e \sim f \iff e = f \vee \exists \text{ cycle containing } e \text{ and } f$$

The equivalence classes are called ***blocks****.*

Note: Two blocks can share at most one vertex, which must be a cut vertex.

Def. 14. ***Block Graph****: Bipartite graph $T = (A \cup B, E_T)$ where: $A = \{\text{cut vertices}\}$ $B = \{\text{blocks}\}$ Edge $\{a, b\} \in E_T \iff a$ is incident to an edge in block b.*

Thm. 15. *If G is connected, the block graph of G is a tree.*

Algorithm 16

```
#DFS & Low Values for cut vertices/bridges in O(|V|+|E|)
low[v] := #smallest DFS number reachable from v by tree edges and <=1 back edge.

low[v] = min{ dfs[v], min_{(v,w) back} dfs[w], min_{(v,w) tree} low[w] }

#A vertex v is a cut vertex iff:
v != root and exists child u with low[u] >= dfs[v].
v = root and has >= 2 children in DFS tree.

#A tree edge (v,w) is a bridge iff:
low[w] >= dfs[v].
```

Eulertours

Def. 17. *An* ***Eulertour*** *is a closed walk in a graph G that contains every edge exactly once.*

Thm. 18. *A connected graph $G = (V, E)$ contains an Eulertour iff the degree of every vertex is even.*

Algorithm 19

```
#Find Eulertour in O(|E|) (Runner and Turtle)
Runner traverses until hitting a visited vertex (cycle found).
Turtle follows path until an unvisited neighbor is found.
Runner is set loose again from there, expanding the tour by cycles.
```

Hamiltonian Cycles

Def. 20. *A* ***Hamiltonian Cycle*** *is a cycle containing every vertex exactly once.*

Thm. 21. *For $m, n \geq 2$, an $n \times m$ grid has a Hamiltonian cycle iff $n \cdot m$ is even.*

Thm. 22. *A bipartite graph $G = (A \cup B, E)$ with $|A| \neq |B|$ cannot contain a Hamiltonian cycle.*

Ex. 23. *The n-dimensional hypercube Q_n is bipartite and has a Hamiltonian cycle iff $n \geq 2$.*

Thm. 24. ***Dirac's Theorem****: Every graph G with $|V| \geq 3$ and minimum degree $\delta(G) \geq |V|/2$ contains a Hamiltonian cycle.*

Algorithm 25

```
#DP for Hamiltonian Cycle O(n^2 * 2^n)
#P[S, x] = 1 if exists 1-x path containing exactly vertices in S.
P[S, x] = max { P[S \setminus \{x\}, y] | y \in N(x) \cap S, y \neq 1 }
```

Travelling Salesman Problem (TSP)

The TSP finds a Hamiltonian cycle in K_n with minimal weight (shortest roundtrip).

Thm. 26. *The TSP is NP-Complete (reducible from Hamiltonian Cycle).*

Metric TSP

For graphs where triangle inequality holds:

$$w(u, v) \leq w(u, w) + w(w, v)$$

There is a 2-approximation algorithm.

Algorithm 27

```
\textbf{2-Approximation for Metric TSP}: 1. Determine MST T. 2. Double edges of T to get Eulerian multigraph T'. 3. Traverse Eulertour, skip already visited vertices.
```

Thm. 28. ***Christofides 1.5-Approximation*** *for Metric TSP: 1. Determine MST T. 2. Find minimal matching M for the odd-degree vertices of T. 3. Determine Eulertour in T with edges of M added.*

Matchings

Def. 29. *An edge set $M \subseteq E$ is a* ***matching*** *in $G = (V, E)$ if no vertex is incident to more than one edge from M.*

- ***Covered****: Vertex v is incident to an edge in M.*
- ***Perfect****: Every vertex is covered ($|M| = |V|/2$).*
- ***Maximal****: Cannot be extended by adding edges.*
- ***Maximum****: No matching with more edges exists.*

Algorithm 30

```
#Greedy Algorithm (Leads to Maximal Matching)
#Size of M is at least half the size of a maximum matching.
M = empty set
While E is not empty:
  Choose arbitrary edge e from E, add to M
  Remove all edges incident to e's endpoints from E
```

Thm. 31. ***Hall's Marriage Theorem****: A bipartite graph $G = (A \cup B, E)$ contains a matching of cardinality $|M| = |A|$ iff:*

$$\forall X \subseteq A : |X| \leq |N(X)|$$

Thm. 32. ***Frobenius Corollary****: Every k-regular bipartite graph contains a perfect matching. In fact, it is a union of perfect matchings. A perfect matching can be found in $O(|E|)$ time if the graph is k- or 2^k-regular.*

Def. 33. *An M-****augmenting path*** *alternates between edges in M and not in M, starting and ending in uncovered vertices.*

Thm. 34. ***Berge's Theorem (1957)****: A matching M is maximum iff there is no M-augmenting path.*

Algorithm 35

```
\textbf{Hopcroft-Karp} ($O(|V|^{1/2}|E|)$ time for bipartite graphs): While there is an augmenting path, find a maximal set of vertex-disjoint shortest augmenting paths using BFS, and augment $M$ along all of them.
```

Coloring

Def. 36. *A* ***k-coloring*** *of $G = (V, E)$ is a mapping $c : V \to [k]$ such that $c(u) \neq c(v)$ for all $\{u, v\} \in E$. A* ***stable set*** *is a set of vertices where no two are adjacent. The* ***chromatic number*** *$\chi(G)$ is the minimum number of colors. $\chi(G) \leq 2 \iff G$ is bipartite.*

Thm. 37. *For $k \geq 3$, the problem "Is $\chi(G) \leq k$?" is NP-complete. For general graphs all ϵ-approximations are NP-hard.*

Algorithm 38

```
\textbf{Greedy Coloring} ($\le \Delta(G) + 1$ colors):
\begin{itemize}
    \item There exists an optimal vertex order yielding exactly $\chi(G)$ colors.
    \item \textbf{Heuristic}: Delete vertices in order of minimum degree. If every subgraph has a vertex of degree $\le k$, this yields $\le k+1$ colors (e.g. trees are $2$-colorable).
\end{itemize}
\textbf{Block-graphs}: If every block can be $k$-colored, the graph can be $k$-colored.
```

Thm. 39. ***Brooks' Theorem**: For a connected graph G that is neither a complete graph K_n nor an odd cycle C_{2n+1}:*

$$\chi(G) \leq \Delta(G)$$

Can be found in $O(|E|)$ time by: 1) If $\Delta(G) = 2$, 2-color. 2) If $\exists v, deg(v) < \Delta(G)$ use greedy heuristic. 3) If $\exists$ cut vertex, color blocks. 4) Else remove non-adj. $x, y \in N(v)$ and color recursively.

Intro: Probability

Def. 40. *A **discrete probability space** is a set of elementary events $\Omega = \{\omega_1, \ldots, \omega_n\}$. Every event ω_i has a probability $P[\omega_i] \in [0,1]$ such that*

$$\sum_{\omega_i \in \Omega} P[\omega_i] = 1$$

Rem. 41. *A finite space where every event has equal probability is a **Laplace-Space**. For any event E:*

$$P[E] = \frac{|E|}{|\Omega|}$$

Def. 42. *An **Event** $E \subseteq \Omega$ has probability $P[E] = \sum_{\omega_i \in E} P[\omega_i]$. The complementary event $\bar{E} = \Omega \setminus E$ has probability $P[\bar{E}] = 1 - P[E]$.*

Thm. 43. *For two events $A, B \subseteq \Omega$: $P[\emptyset] = 0$, $P[\Omega] = 1$, $P[A] \in [0,1]$. If $A \subset B$ then $P[A] \leq P[B]$.*

Thm. 44. ***Union Bound**: For events $A_1, \ldots, A_n$:*

$$P[\bigcup_{i=1}^{n} A_i] \leq \sum_{i=1}^{n} P[A_i]$$

(Equality iff A_i are pairwise disjoint).

Thm. 45. ***Inclusion-Exclusion Principle**: For $n \geq 2$:*

$$P[\bigcup_{i=1}^{n} A_i] = \sum_{k=1}^{n} (-1)^{k+1} \sum_{I \subset [n], |I|=k} P[\bigcap_{j \in I} A_j]$$

Conditional Probability

Def. 46. *For $P[B] > 0$, the **conditional probability** of A given B is:*

$$P[A \mid B] = \frac{P[A \cap B]}{P[B]}$$

Rem. 47. *Always holds: $P[A \mid \Omega] = P[A]$, $P[A \mid A] = 1$, $P[A \mid \bar{A}] = 0$, $P[A \cap B] = P[A \mid B]P[B]$.*

Thm. 48. ***Multiplication-Theorem**: If $P[\bigcap_{i=1}^{n-1} A_i] > 0$:*

$$P[\bigcap_{i=1}^{n} A_i] = P[A_1] \cdot P[A_2 \mid A_1]$$

$$\cdots\cdot P[A_n \mid \bigcap_{i=1}^{n-1} A_i]$$

Thm. 49. ***Theorem of total probability**: For pairwise disjoint A_i and $B \subseteq \bigcup_{i=1}^{n} A_i$:*

$$P[B] = \sum_{i=1}^{n} P[B \mid A_i] P[A_i]$$

Thm. 50. ***Bayes' Theorem**: For exactly defined disjoint A_i, $B \subseteq \bigcup A_i$, and $P[B] > 0$:*

$$P[A_i \mid B] = \frac{P[B \mid A_i] P[A_i]}{\sum_{j=1}^{n} P[B \mid A_j] P[A_j]}$$

Independence

Def. 51. *Two events A, B are **independent** if:*

$$P[A \cap B] = P[A]P[B] \text{ or } P[A \mid B] = P[A]$$

Events $A_1, \ldots, A_n$ are independent if for all $I \subseteq \{1, \ldots, n\}$:

$$P[\bigcap_{i \in I} A_i] = \prod_{i \in I} P[A_i]$$

Lem. 52. *Let $A_1, \ldots, A_n$ be events, $A_i^0 = \bar{A}_i$, $A_i^1 = A_i$. They are independent iff $\forall (s_1, \ldots, s_n) \in \{0,1\}^n$:*

$$P[\bigcap_{i=1}^{n} A_i^{s_i}] = \prod_{i=1}^{n} P[A_i^{s_i}]$$

Lem. 53. *If A, B, C are independent, then so are $A \cap B$ and C, and $A \cup B$ and C.*

Random Variables

Def. 54. *A **random variable** is a map $X : \Omega \to \mathbb{R}$. The **probability mass function (PMF)** is $f_X(x) = P(X = x)$. The **distribution function (CDF)** is $F_X(x) = P(X \leq x)$.*

Expectation

Def. 55. *The expectation of X is defined as:*

$$\mathbb{E}[X] := \sum_{\alpha \in W_x} \alpha P(X = \alpha)$$

Thm. 56. *For a random variable X with values in $\mathbb{N}_0$:*

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} P(X \geq k)$$

Def. 57. *Indicator-variable X_E for $E \subseteq \Omega$: $X_E(\omega) = 1$ if $\omega \in E$, else 0.*

$$\mathbb{E}[X_E] = P(E)$$

Thm. 58. ***Linearity of Expectation**: For random variables $X_1, \ldots, X_n$ and $a_1, \ldots, a_n, b \in \mathbb{R}$:*

$$\mathbb{E}[\sum_{i=1}^{n} a_i X_i + b] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i] + b$$

Algorithm 59

```
#Maximal Stable Set Algorithm
S[v] = 1/0 with probability p
For all {u,v} in E:
  if S[u] == 1 and S[v] == 1: S[v] = 0
Expected size: $\mathbb{E}[\mid S \mid] \geq n*p - m*p^2$
```

Variance

Def. 60. *For X with $\mathbb{E}[X] = \mu$, the **variance** is:*

$$Var[X] := \mathbb{E}[(X - \mu)^2]$$

Standard deviation $\sigma := \sqrt{Var[X]}$.

Thm. 61.

$$Var[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Lem. 62. *For $a, b \in \mathbb{R}$: $Var[aX + b] = a^2 Var[X]$*

Def. 63. *$\mathbb{E}[X^k]$ is the k-th **moment**. $\mathbb{E}[(X - \mathbb{E}[X])^k]$ is the k-th **central moment**.*

Multiple & Combined Variables

Def. 64. *RVs $X_1, \ldots, X_n$ are **independent** iff $\forall a_i \in W_{X_i}$:*

$$P[X_1 = a_1, \ldots, X_n = a_n] = \prod_{i=1}^{n} P[X_i = a_i]$$

Thm. 65. *If X, Y are independent, $f_{X+Y}(\alpha) = \sum_\beta f_X(\beta) f_Y(\alpha - \beta)$.*

Rem. 66. ***Calculation Rules**:*

- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
- *If X, Y independent: $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$*
- *If X, Y independent: $Var[X + Y] = Var[X] + Var[Y]$*

Note $Var[XY] \neq Var[X]Var[Y]$ in general.

Thm. 67. ***Total Expectation**: For pairwise disjoint events $A_1, \ldots, A_n$ covering Ω:*

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X \mid A_i] P[A_i]$$

Thm. 68. ***Wald's Equation**: If N and X are independent RVs and $Z := \sum_{i=1}^{N} X_i$ (where X_i are independent copies of X):*

$$\mathbb{E}[Z] = \mathbb{E}[N]\mathbb{E}[X]$$

Distributions

Def. 69. ***Bernoulli(p)**: $X \in \{0, 1\}$*

$$P[X = 1] = p \quad P[X = 0] = 1 - p$$

$$\mathbb{E}[X] = p \quad Var[X] = p(1 - p)$$

Def. 70. ***Binomial(n, p)**: $X \in \{0, \ldots, n\}$*

$$P[X = k] = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\mathbb{E}[X] = np \quad Var[X] = np(1 - p)$$

Def. 71. ***Poisson(λ):***

$$P[X = k] = \frac{e^{-\lambda}\lambda^k}{k!}$$
$$\mathbb{E}[X] = \lambda \quad Var[X] = \lambda$$

Def. 72. ***Geometric(p):***

$$P[X = k] = p(1-p)^{k-1}$$
$$\mathbb{E}[X] = \frac{1}{p} \quad Var[X] = \frac{1-p}{p^2}$$

(Number of throws until first success).

Def. 73. ***Negative Binomial of order n:***

$$P[X = k] = \binom{k-1}{n-1} p^n (1-p)^{k-n}$$

(Number of throws until n-th success).

Bounding Probability

Thm. 74. ***Markov's Inequality:*** *For non-negative RV X and $t \geq 0$:*

$$P[X \geq t] \leq \frac{\mathbb{E}[X]}{t} \quad \text{or} \quad P[X \geq t\mathbb{E}[X]] \leq \frac{1}{t}$$

Thm. 75. ***Chebyshev's Inequality:*** *For any RV X and $t \geq 0$:*

$$P[|X - \mathbb{E}[X]| \geq t] \leq \frac{Var[X]}{t^2}$$

Chernoff Bounds

Applies to sums of independent Bernoulli RVs $X = \sum X_i$ with $P[X_i = 1] = p_i$.

Thm. 76. *For $0 \leq \delta \leq 1$:*

$$P[X \geq (1+\delta)\mathbb{E}[X]] \leq e^{-\frac{1}{3}\delta^2\mathbb{E}[X]}$$
$$P[X \leq (1-\delta)\mathbb{E}[X]] \leq e^{-\frac{1}{2}\delta^2\mathbb{E}[X]}$$

For larger deviations ($t \geq 2e\mathbb{E}[X]$):

$$P[X \geq t] \leq 2^{-t}$$

Target Shooting (Sampling)

To estimate $|S|/|U|$, sample n elements uniformly. Let $Y = \frac{1}{n}\sum Y_i$ be the fraction of samples in S.

Thm. 77. *With given $\delta, \epsilon \geq 0$, if $n \geq \frac{3|U|}{|S|\epsilon^2} \ln \frac{2}{\delta}$, then*

$$Y \in \left[(1-\epsilon)\frac{|S|}{|U|}, (1+\epsilon)\frac{|S|}{|U|}\right]$$

with probability $\geq 1 - \delta$.